JACOBUS PHILIP HAUPT

Paris, France ◇ +33 7 44 98 61 45 ◇ jphaupt@proton.me
jphaupt.github.io ◇ GitHub ◇ LinkedIn ◇ Google Scholar

SUMMARY

Computational scientist (PhD, Max Planck Institute) with five years of research experience in Monte Carlo methods, stochastic optimisation and large-scale numerical computing, currently working on machine-learning foundation models. Taking algorithms from mathematical derivation to tested implementations in Python, Fortran and C++ on HPC clusters. Building finance foundations through independent projects and Coursera's Financial engineering and Risk Management Specialization. Seeking quantitative researcher and quantitative trader roles applying statistical modelling and simulation to financial markets.

RESEARCH EXPERIENCE

Postdoctoral Researcher January 2026 – Present
Sorbonne University *Paris, France*

- Combining selected configuration interaction (an iterative sparse eigensolver) with diffusion Monte Carlo to generate high-accuracy training data for machine-learning foundation models of molecules for drug discovery
- Building ML training and evaluation pipelines in JAX and PyTorch; HPC development in Python and C++; developing workflows on the Aurora supercomputer
- Industrial and national-lab collaborations with Qubit Pharmaceuticals and Argonne National Lab

Doctoral Researcher, then Postdoctoral Researcher November 2020 – November 2025
Max Planck Institute for Solid State Research *Stuttgart, Germany*

- Developed stochastic algorithms for high-dimensional eigenvalue problems (arising in *ab initio* quantum chemistry), from mathematical derivation through to parallel implementation and publication
- Lead developer of PyTCHInt, a Fortran numerical-integration library with a Cython/Python interface; introduced CI/CD, unit testing and auto-generated documentation
- Co-lead developer of NECI, the largest MPI-parallel Fortran code for full configuration interaction quantum Monte Carlo, a stochastic sparse-matrix eigensolver; introduced various improvements for transcorrelated methods, including a variance reduction method
- Designed and optimised many-parameter models under Monte Carlo noise using variational and stochastic optimisation (first author, *J. Chem. Phys.* 2023)
- Statistical analysis of correlated Monte Carlo time series: error estimation via blocking analysis, bias and variance control, convergence diagnostics

SELECTED PERSONAL PROJECTS

CIPSIpy JAX, LLMs

- Implemented a selected configuration interaction solver from scratch in JAX: iterative subspace selection with perturbative importance ranking and sparse diagonalisation
- Benchmarked against reference results and fully unit-tested; methods and results documented online

Biotech Event Study Python, pandas, yfinance

- Proof-of-concept event study of biotech equity returns around FDA approval decisions using cumulative abnormal returns
- Found larger price reactions to unanticipated approvals than to expected ones

EDUCATION

PhD in Computational Chemistry 2020 – 2025
Max Planck Institute for Solid State Research / University of Stuttgart *Stuttgart, Germany*

- Thesis awarded *magna cum laude*: *Development of the Transcorrelated Full Configuration Interaction Quantum Monte Carlo Method*; advisor Prof. Ali Alavi

Doctoral Studies in Theoretical Physics 2019 – 2020
University of Toronto *Toronto, Canada*

- Direct-entry PhD scholarship; graduate coursework and research in computational electromagnetics

BSc in Combined Honours Physics & Mathematics, Minor in Computer Science
With Co-Operative Education 2014 – 2019
University of British Columbia *Vancouver, Canada*

- GPA 85.8/100 (4.00), graduated with distinction; graduate-level coursework in mathematics and physics; advanced undergraduate coursework in computer science
- UBC Science Scholar, Palliser-Wilson Scholarship in Mathematics, Dean's Honour List (all years)

EARLIER EXPERIENCE

- **Research Intern, ML in atomic physics** (UBC, 2019): sparse regression (lasso) for model selection in quantum scattering; NSERC USRA award
- **Research Intern, ML in biophysics** (EPFL, 2018): recurrent autoencoders (Keras/TensorFlow) to decode polymer sequences from noisy nanopore current signals
- **Research Intern, ML for experiment automation** (MPI Stuttgart, 2017): neural networks to pre-screen scanning tunnelling spectroscopy data
- **Business Intelligence Analyst** (Royal Canadian Mounted Police, 2016): SQL and queueing-theory analysis of province-wide staffing data to benchmark detachment performance; initiated a collaboration with UBC

SKILLS

- **Programming:** Python, Fortran (advanced); C++, Julia, Bash (intermediate); SQL, R, MATLAB
- **Libraries and tools:** NumPy, JAX, PyTorch, pandas, Cython, MPI, OpenMP, CUDA, git, CI/CD
- **Quantitative methods:** Monte Carlo, variance reduction, stochastic optimisation, finite-difference and finite-element methods, numerical linear algebra, Bayesian methods, regularised regression, neural networks
- **Training:** HLRS Stuttgart courses in MPI/OpenMP, modern C++ design, CUDA and Julia HPC; Coursera *Financial Engineering and Risk Management* Specialization [in progress, certificates so far displayed on LinkedIn]
- **Languages:** English (native), German (B2), French (A1)

SELECTED PUBLICATIONS

Six peer-reviewed articles, three as first author, two more currently being drafted (see Google Scholar for up-to-date list), including:

- **Haupt** *et al.*, Optimizing Jastrow factors for the transcorrelated method, *J. Chem. Phys.* **158** (2023)
- **Haupt** *et al.*, Transcorrelated methods for multireference problems, *J. Chem. Phys.* **163** (2025)